

1. Lab Overview

In this lab, I practiced creating and managing shared folders in a Windows Server 2019 domain environment. The lab focused on creating a shared folder, accessing it from a Windows 10 client, configuring share permissions, configuring NTFS permissions, and testing how permissions affect user access.

This lab builds on the previous Active Directory labs by applying user and group management to real file access scenarios. These are common tasks for IT Support Analysts, Help Desk Analysts, Desktop Support Technicians, and Junior System Administrators.

2. Business Scenario

This lab simulates a business environment where users need access to shared folders on a network server. As the system administrator, I created a shared folder, configured access permissions, and tested user access from a domain-joined Windows 10 client.

The goal was to understand how share permissions and NTFS permissions work together to control access to network resources. Your original lab section lists the goals as creating, publishing, and accessing a shared folder, configuring and testing share permissions, and configuring and testing NTFS permissions.

3. Lab Objectives

The objectives of this lab were to:

- Create a folder on Windows Server 2019.
- Share the folder over the network.
- Access the shared folder from a Windows 10 client.
- Configure share permissions.
- Configure NTFS permissions.
- Test access using different users or groups.
- Understand the difference between share permissions and NTFS permissions.
- Troubleshoot permission conflicts.
- Apply the principle of least privilege to shared resources.

4. Lab Environment

Item	Details
Server VM	Windows Server 2019

Client VM	Windows 10
Domain	wara.edu
Main Tool	File Explorer
Other Tools	Active Directory Users and Computers, Server Manager
Permission Types	Share permissions and NTFS permissions
Main Concept	Most restrictive permission applies during remote access

5. Tools Used

- Windows Server 2019
- Windows 10
- File Explorer
- Active Directory Users and Computers
- Shared folder properties
- Advanced Sharing
- Share permissions
- NTFS Security tab
- Domain users and groups

6. Steps Performed

Step 1: Created a Folder for Sharing

I created a folder on the Windows Server 2019 VM to use as a network shared folder. This folder was used to test how users access resources over the network.

Creating a shared folder is a common task in business environments because users often need shared access to department files, project folders, or network resources.

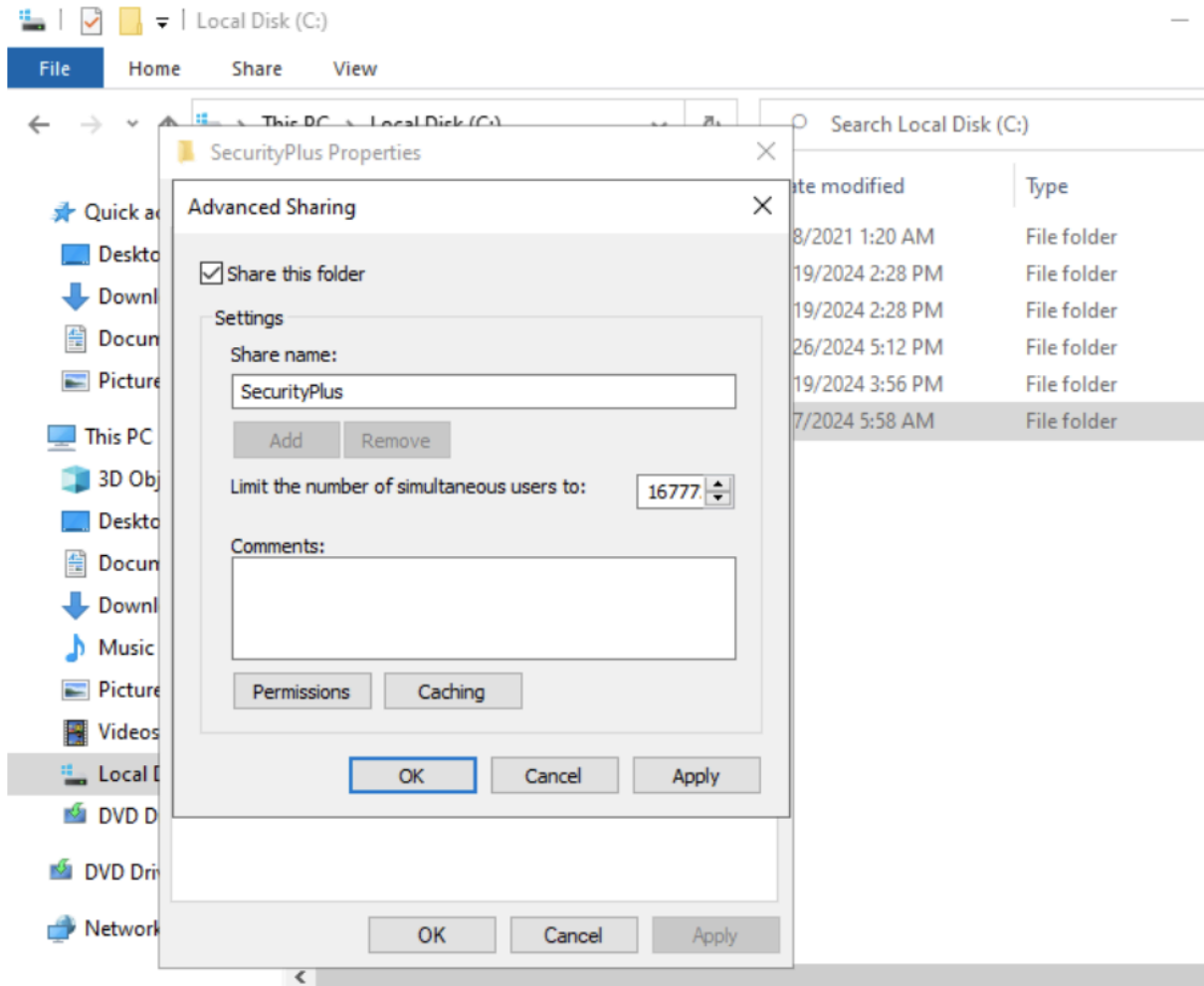


Figure 1: SecurityPlus folder created on Windows Server 2019 for network sharing.

Step 2: Enabled Folder Sharing

Next, I opened the folder properties and enabled sharing. I used the sharing options to make the folder available over the network.

This allowed the folder to be accessed from the Windows 10 client using a network path.

Example network path format:

\\WARA-SERVER\SharedFolderName

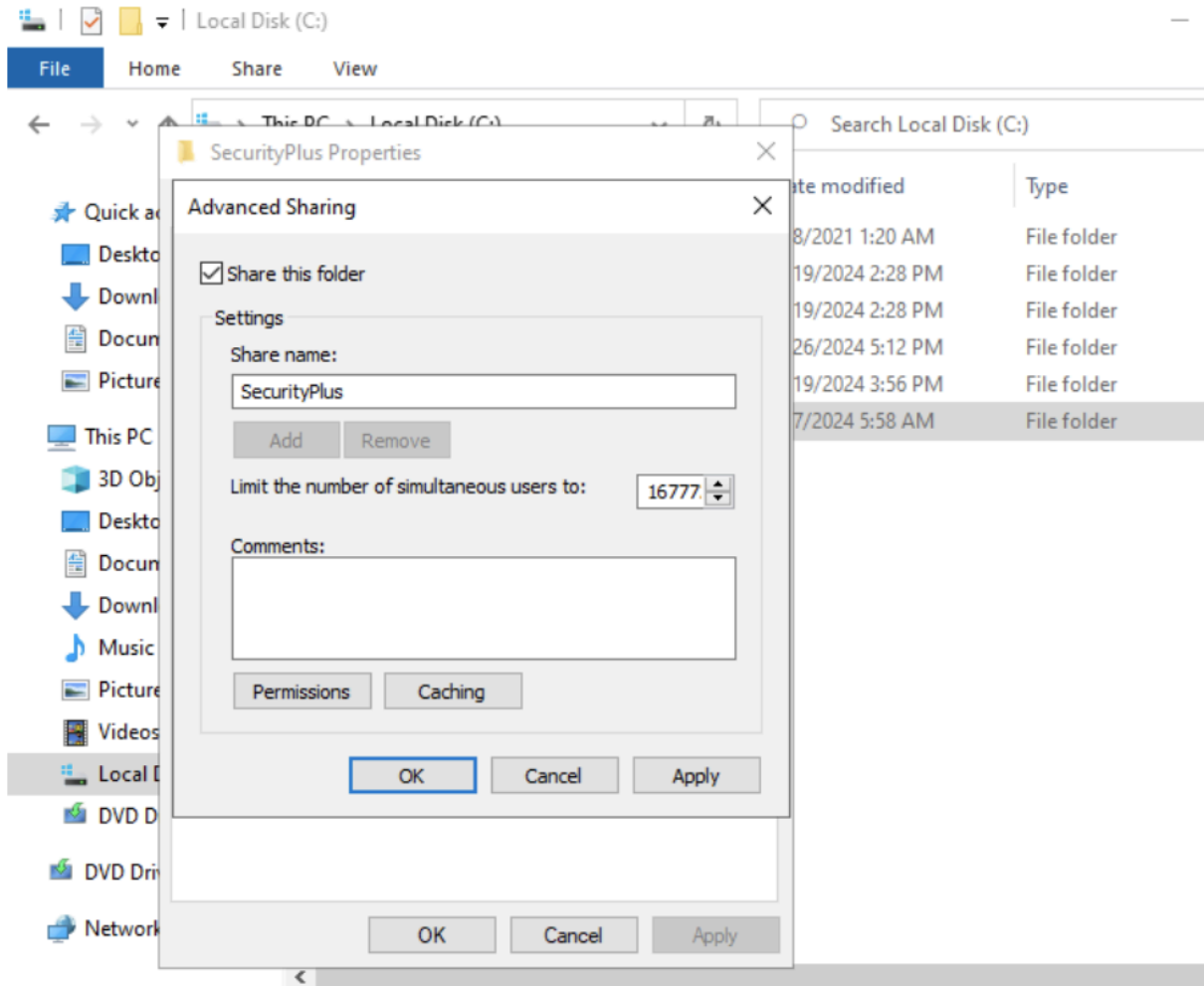


Figure 2: Advanced Sharing enabled for the SecurityPlus folder.

Step 3: Configured Share Permissions

After enabling sharing, I reviewed and configured the **share permissions**.

Share permissions control access to the folder when users connect to it over the network. Common share permission levels include:

Read
Change
Full Control

The goal was to control what users or groups could do when accessing the shared folder remotely.

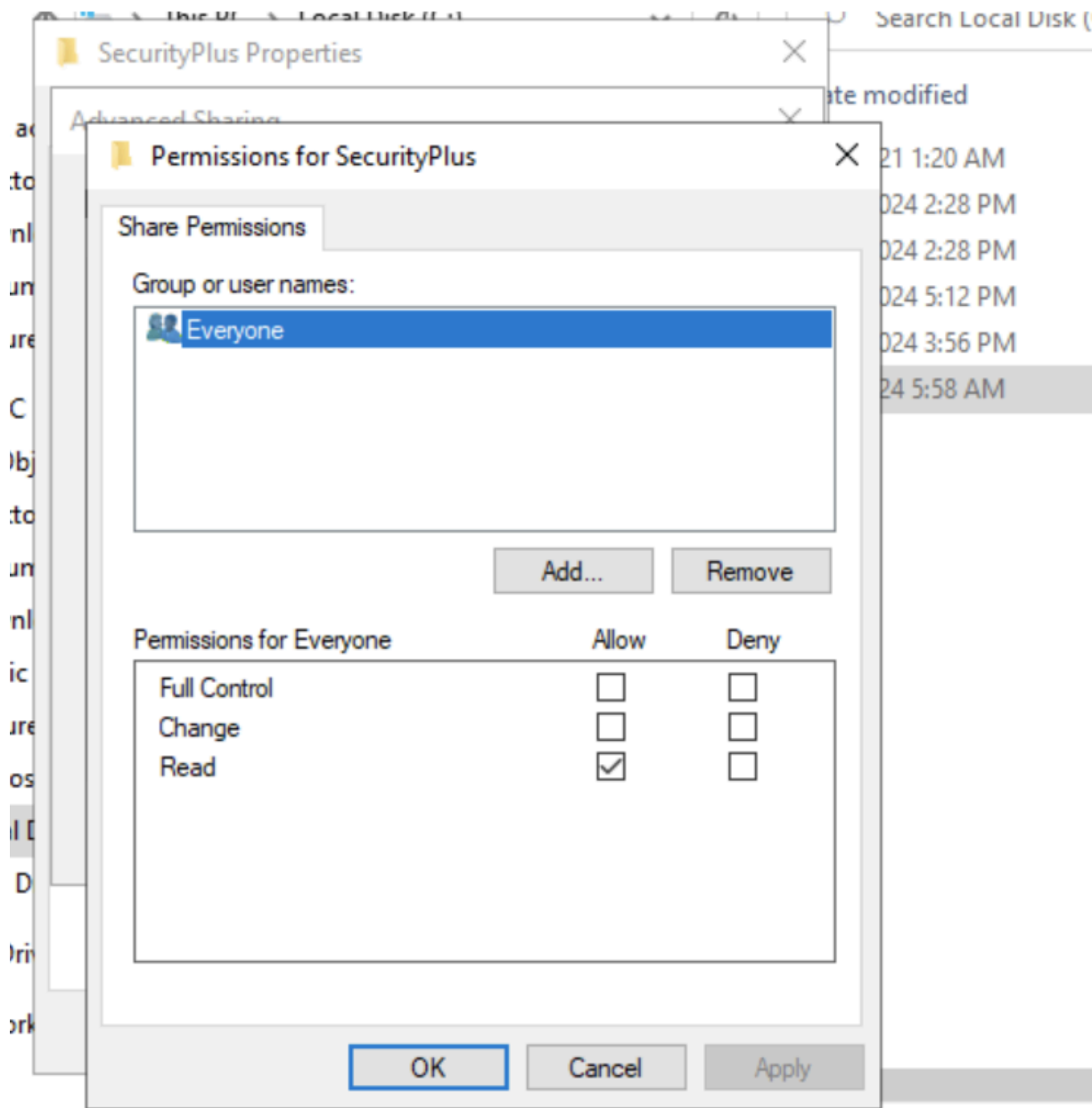


Figure 4: Share permissions configured with Read access for Everyone.

Step 4: Configured NTFS Permissions

Next, I configured **NTFS permissions** from the folder's **Security** tab.

NTFS permissions control access to the folder at the file system level. These permissions apply whether the user is accessing the folder locally or remotely.

Common NTFS permissions include:

Read & Execute
List Folder Contents
Read
Write
Modify
Full Control

Your lab notes show that the Security tab was used for NTFS permission configuration, and the permissions list displayed access such as Read & Execute, List Folder Contents, Read, and Special Permissions.

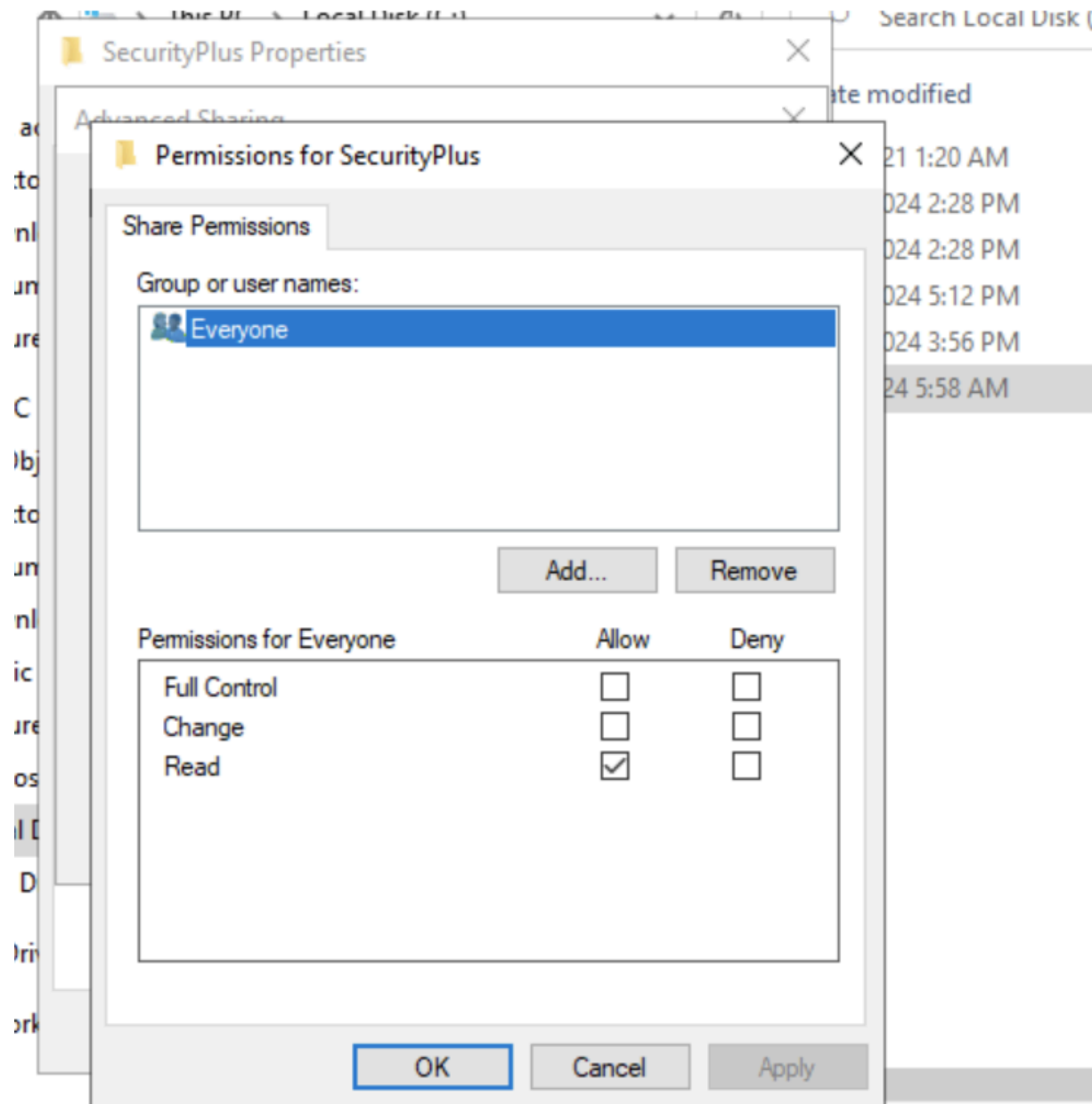


Figure 4: NTFS permissions reviewed and configured from the Security tab.

Step 5: Tested Access from the Windows 10 Client

After configuring the shared folder, I tested access from the Windows 10 client VM.

I used the network path to access the shared folder and verified whether the user had the correct level of access.

This step helped confirm whether the permission configuration worked as expected.

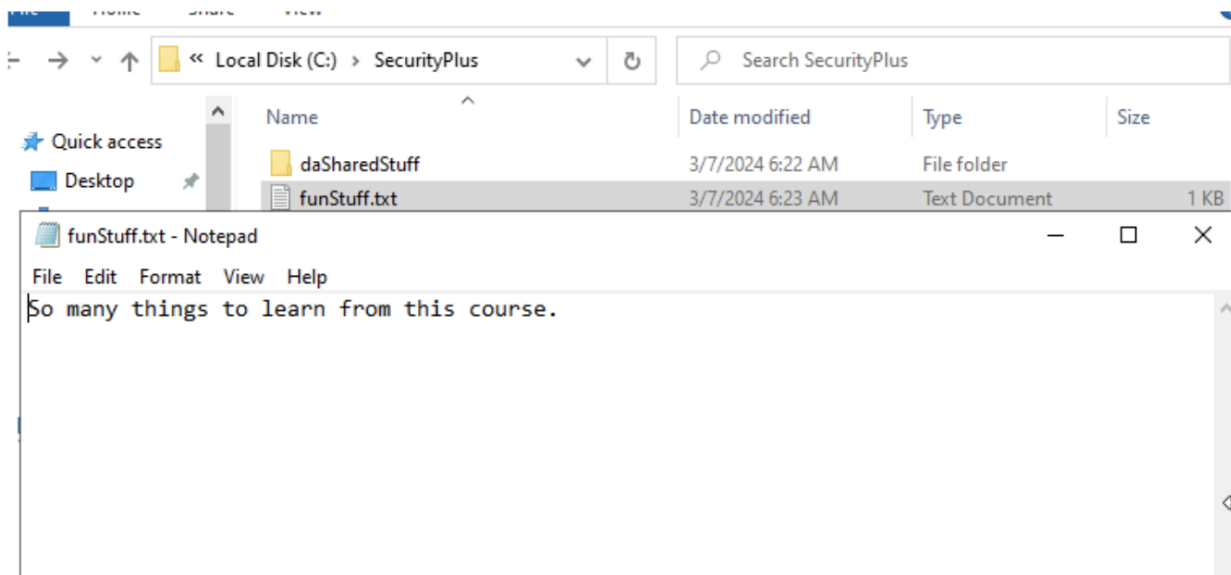


Figure 5: Shared folder accessed from the Windows 10 domain client.

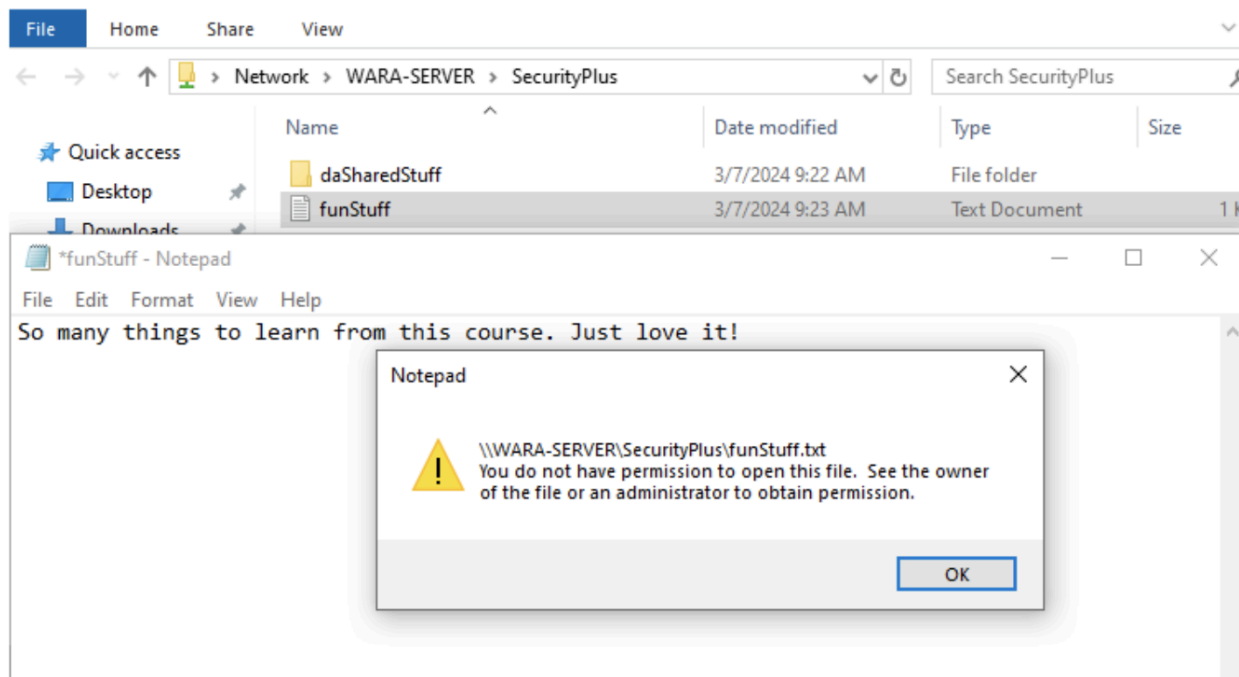
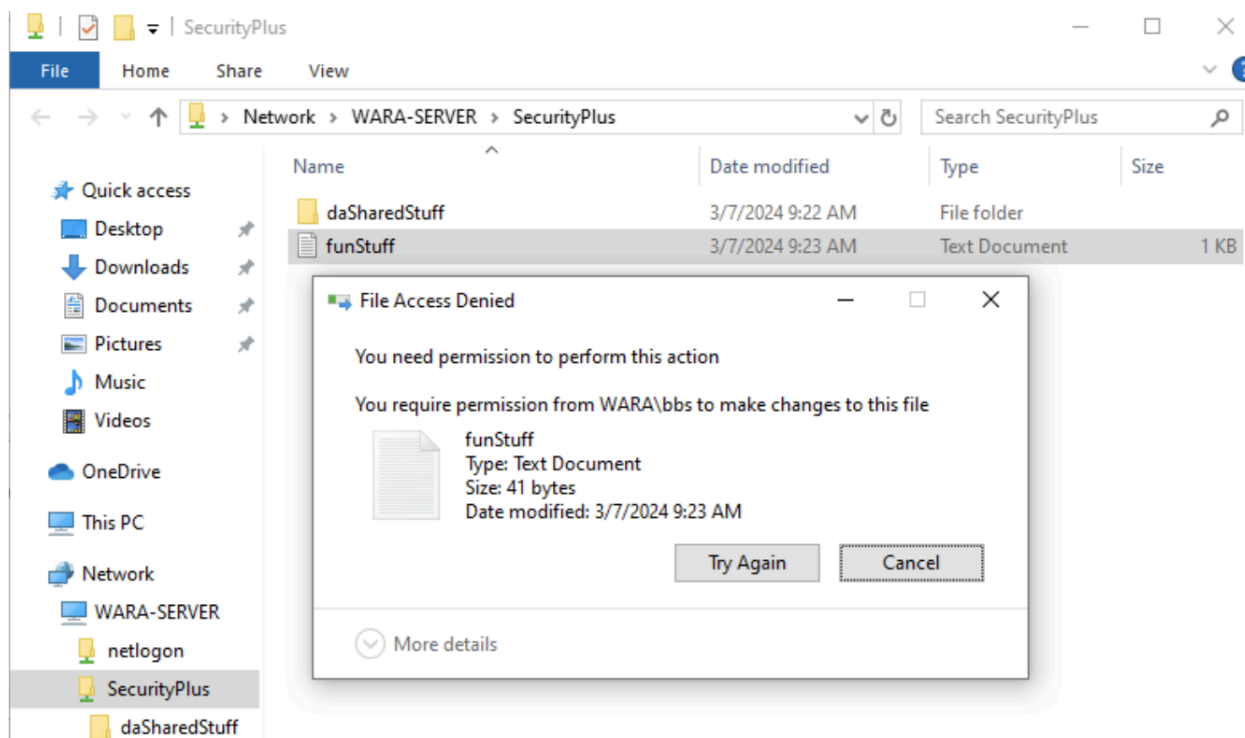
Step 6: Tested Permission Behavior

I tested how different permission settings affected user access.

This included checking whether a user could:

- Open the shared folder
- View folder contents
- Create files
- Modify files
- Delete files
- Access denied messages when permissions were too restrictive

This helped me understand the relationship between share permissions and NTFS permissions.



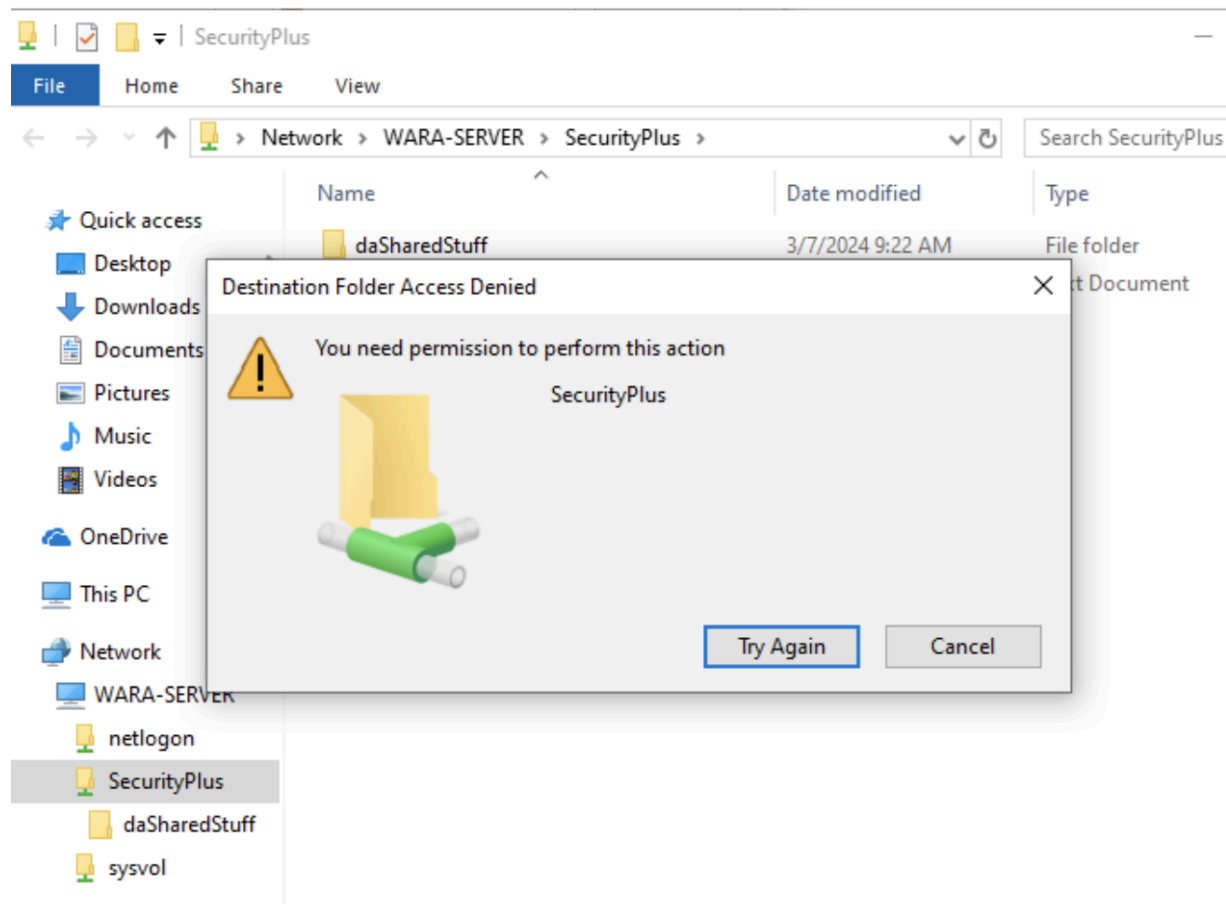


Figure 6: Access denied message showing the effect of share and NTFS permission settings.

Step 7: Analyzed Share vs. NTFS Permission Conflicts

One of the most important lessons from this lab was that when share permissions and NTFS permissions conflict during remote access, the **more restrictive permission applies**.

For example:

Share Permission	NTFS Permission	Effective Remote Access
Full Control	Read	Read
Read	Full Control	Read

Change	Modify	Change/Modify depending on restriction
Full Control	Full Control	Full Control

Your lab analysis explains that share permissions control access over the network, NTFS permissions control access to the file system, and when there is a conflict, Windows applies the most restrictive permission.

7. Key Findings

Finding	Explanation
Shared folders provide network access	Users can access server folders from a domain client using a UNC path.
Share permissions control remote access	Share permissions apply when users access the folder over the network.
NTFS permissions control file system access	NTFS permissions apply locally and remotely.
Effective access depends on both permission types	Remote access uses the most restrictive combination of share and NTFS permissions.
Groups are better than individual users	Assigning permissions to groups is easier to manage and more scalable.
Least privilege is important	Users should only receive the access they need to perform their work.

8. Analysis

This lab showed that file sharing in Windows is not only about creating a shared folder. It also requires careful permission planning.

The most important concept is that share permissions and NTFS permissions work together. A user may appear to have access from one permission area, but still be blocked by the other. This is why IT support professionals must check both the **Sharing** tab and the **Security** tab when troubleshooting folder access issues.

This lab also showed why permission management should be group-based. Instead of giving permissions to each user individually, it is better to create groups and assign permissions to those groups. This makes access control easier to manage and reduces mistakes.

9. IT Support Relevance

This lab is highly relevant to IT Support Analyst and Help Desk roles.

It connects to real job tasks such as:

- Creating shared folders
- Troubleshooting “access denied” errors
- Checking share permissions
- Checking NTFS permissions
- Adding users to access groups
- Testing access from a client computer
- Understanding effective permissions
- Supporting department file shares
- Applying least privilege
- Documenting access changes

10. Skills Demonstrated

This lab demonstrates:

- Windows Server 2019 file sharing
- Shared folder creation
- UNC path access
- Share permission configuration
- NTFS permission configuration
- Access control troubleshooting
- Group-based permissions
- Least privilege
- Domain client testing
- IT support documentation

11. Ticket-Style Summary

Issue:

A shared folder needed to be created and secured so domain users could access it with the correct permissions.

Impact:

Without correct share and NTFS permissions, users may either be blocked from required resources or receive more access than needed.

Steps Taken:

- Created a folder on Windows Server 2019.
- Enabled sharing on the folder.
- Configured share permissions.
- Reviewed and configured NTFS permissions.
- Accessed the shared folder from the Windows 10 client.
- Tested user access.
- Compared share permissions and NTFS permissions.
- Identified that the most restrictive permission applies during remote access.

Root Cause:

Shared folder access depends on both network-level share permissions and file system-level NTFS permissions.

Resolution:

Configured the shared folder with appropriate share and NTFS permissions, then tested access from the domain client.

Status:

Completed successfully.

12. Lessons Learned

In this lab, I learned that Windows file sharing requires both share permissions and NTFS permissions to be configured correctly. I also learned that when these permissions conflict, the most restrictive permission applies.

This is important for IT support because many user access issues happen when someone is missing group membership, share permissions, or NTFS permissions. I also learned that using groups for permissions is better than assigning access to individual users.